



STINK3024 Machine Learning

Session I 2025/2026 (A252)

Assignment 2 Machine Learning Model Development (20%)

LOC 3a

Type: Group work (4 members per group – as determined)

Release date: Week 8

Submission: Week 10

Assignment Goals

By the end of this assignment, students should be able to:

1. Select and justify an appropriate machine learning algorithm based on the nature of the dataset and problem domain (e.g., classification, regression, clustering).
2. Develop and train a machine learning model using suitable tools or programming environments.
3. Apply data preprocessing and feature engineering techniques to enhance model performance and accuracy.
4. Evaluate model performance using appropriate metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.
5. Compare multiple models or parameter settings to determine the most effective approach for the given problem.
6. Interpret and communicate model results through clear visualizations and evidence-based discussion.
7. Demonstrate good practices in model validation, including data splitting (training/testing) and cross-validation techniques.

Term and Condition: A penalty of 2 marks will be applied for plagiarism or late submission

Objective: To expose student to the development of machine learning techniques on various domains

Instructions:

You are required to:

1. **Split the Dataset**
 - Divide the dataset into training and testing sets using appropriate methods (e.g., 70/30 split, stratified sampling, cross validation).
2. **Model Selection**
 - Choose at least three different machine learning algorithms such as:
 - Linear Models (e.g., Linear Regression, Logistic Regression)

- Tree-Based Models (e.g., Decision Trees, Random Forests, XGBoost)
 - Support Vector Machines
 - K-Nearest Neighbors
 - Neural Networks (basic MLP or using a framework like TensorFlow/Keras or PyTorch)
 - Deep Learning
 - Justify your choice of algorithms.
3. **Model Training**
- Train the models on the training dataset.
 - Tune hyperparameters using methods such as grid search or randomized search.
4. **Model Evaluation**
- Evaluate the models using appropriate metrics (accuracy, precision, recall, F1-score, ROC-AUC, RMSE, etc. depending on the problem type: classification or regression).
 - Use cross-validation to ensure robustness of results.
5. **Model Comparison**
- Compare model performances and identify the best-performing model.
 - Provide visualizations (e.g., confusion matrix, ROC curve, learning curves).
6. **Conclusion**
- Summarize findings.
 - Discuss limitations and potential improvements.
 - Reflect on the challenges faced during model development.